



**Verified Carbon
Standard**

GUANGDONG LANKOU 26MW HYDRO POWER PROJECT

Document Prepared by Beijing 3050 Technology Co., Ltd

Project Title	Guangdong Lankou 26MW Hydro Power Project
Version	02
Report ID	1 st
Date of Issue	14/07/2022
Project ID	371
Monitoring Period	01/06/2012 to 31/10/2017
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1 PROJECT DETAILS

1.1 Summary Description of the Implementation Status of the Project

Guangdong Lankou 26MW Hydro Power Project (hereinafter referred to as “the project”) involves the construction and operation of a hydro power station (with a new reservoir) which is developed and owned by Shenzhen City Fuyuan Industry Group Co., Ltd. The project is located on the Dongjiang River which is one of the major four rivers of Zhu Jiang in Bainitang Village, Lankou Town, Dongyuan County, Heyuan City, Guangdong Province, China.

The total installed capacity of the proposed project is 26MW(3*8.66MW), the surface area of the reservoir at full level is 141,441m² thus the power density of the project is 183.82W/m². On the average, the project activity is expected to operate 3,462hours per year and the net grid-connected electricity generation of 80,070MWh. The power generated would be transmitted to Heyuan City Grid via the 110kV Lankou transformer substation, then to the Guangdong Provincial Grid, and finally to the China Southern power Grid (CSPG).

The construction of the project was started on 04/12/2005, 3 units of 8.66MW turbine and generator sets were commissioned on 01/11/2007, 01/01/2008, 01/05/2008 respectively.

The project has operated without any accidental or emergency events that might affect the accuracy and/or implementation of the monitoring activities. The net power supply during this monitoring period (01/06/2012-31/10/2017) is 420,237.338MWh. The total emission reductions claimed in this monitoring period are 366,109tCO₂e.

1.2 Sectoral Scope and Project Type

Sectoral Scope 1: Energy industries (renewable /non-renewable sources);

Project type: renewable energy project;

The project is not a grouped project.

1.3 Project Proponent

Organization name	Shenzhen City Fuyuan Industry Group Co., Ltd
Contact person	Zhang Defu

Title	Director
Address	460 Fanshen Road, district 45, Bao'an District, Shenzhen City,China
Telephone	+86-0755-27578898
Email	1477209050@qq.com

1.4 Other Entities Involved in the Project

Organization name	Beijing 3050 Technology Co.,Ltd
Role in the Project	Consultant
Contact person	Zheng Jiangzhuo
Title	Project Manager
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1.5 Project Start Date

01/11/2007(the commissioning of the 1st turbine and generator)

1.6 Project Crediting Period

Crediting period length: 01/11/2007 - 31/10/2017

Crediting period type: 10-year renewable (renewable twice)

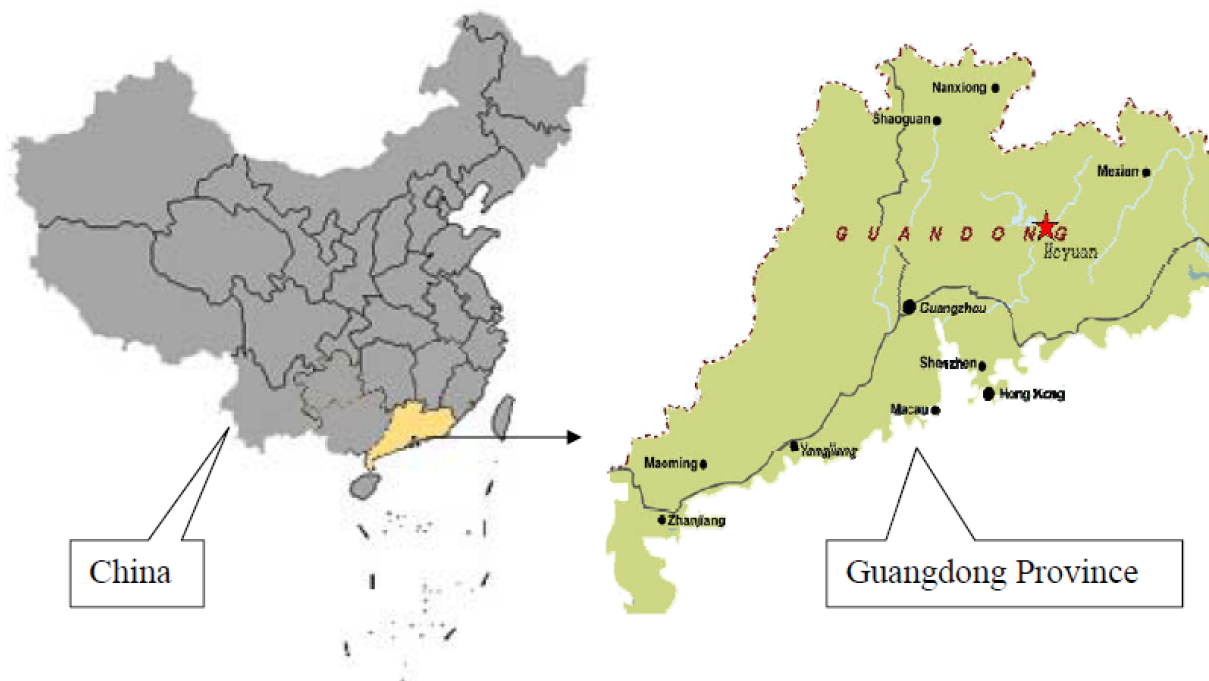
There is a deviation of the project crediting period. The project is registered under VCS version 2007 and completed validation before 19/03/2020, Thus, it remains eligible to apply the crediting period requirements under VCS standard Version 3 as per the appendix 2 of VCS standard of VCS standard Version 4.1 which shall be a maximum of ten years and maybe renewed at most twice, so the first renewable crediting period of the project should be updated from 01/11/2007~06/07/2009 to 01/11/2007 - 31/10/2017 (ten years).

According to VCS standard, the renewal of the VCS crediting periods could not be more than the end of the 3*7 years of CDM crediting period, that is 31/10/2028. As per VCS standard version 4.1, 3.8.9 (3) “the project shall be validated against the (current) scope of the VCS. Such validation report shall be issued after the end of the (previous) project crediting period but within two years after the end of the (previous) project crediting period”, however as the project did not make a timely renewal of the 2nd VCS crediting period, so the project is only eligible for the first VCS crediting period of ten years and ends on 31/10/2017.

1.7 Project Location

The project is located at the Bainitang village, Lankou Town, Dongyuan County. It is 5km from the centre of Lankou Town, 42.9km away from the centre of Heyuan City. And its geographical coordinates are east-longitude 115°02′ 44′ ′ and north-latitude 23°55′ 56′ ′ .

Figure 1 below shows the location of the project.



1.8 Title and Reference of Methodology

Methodology:

Approved consolidated baseline methodology ACM0002 “Consolidated baseline methodology for grid-connected electricity generation from renewable sources” (version

07)

Reference:

<https://cdm.unfccc.int/methodologies/DB/XP2LKUSA61DKUQC0PIWPGWDN8ED5PG>

Tools applied:

“Tool for the Demonstration and Assessment of Additionality” (version 5.2)

“Tool to calculate the emission factor for an electricity system” (version 01.1)

1.9 Participation under other GHG Programs

The project was registered with UNFCCC as a Clean Development Mechanism (CDM) project (reference number: 2458) with the 21 (3*7) years crediting period started from 07/07/2009. Please refer to the following link for details:

<https://cdm.unfccc.int/Projects/DB/JCI1238492309.45/view>

Except for CDM, the project did not register under any other GHG program.

The emission reductions generated from the project from 01/11/2007 to 31/05/2012 were issued as the following credits as shown in the table below.

Start Date	End Date	GHG Emission Reductions (tCO _{2e})	Credit	Program
01/11/2007	06/07/2009	82,009	VCU	VCS
07/07/2009	31/03/2010	29,752	CER	CDM
01/04/2010	31/05/2011	77,676	CER	CDM
01/06/2011	31/05/2012	58,868	CER	CDM

The emission reductions generated during this monitoring period, from 01/06/2012 to 31/10/2017 (the first and last days included) will request for VCUs and will not be submitted for any other emission reduction certificate or carbon offset. Hence, it is confirmed that the emission reductions will not be double counted.

1.10 Other Forms of Credit

Emission Trading Programs and other Binding Limits

China launched a national emissions trading scheme at 16/07/2021 only covering the fossil fuel fired power industry. Other regional mandatory pilot emissions trading schemes which operated earlier in China include carbon intensive industries such as thermal power generation, petrochemical, chemical, building materials, iron and steel, non-ferrous, paper, aviation etc.,

not including renewable energy sector¹, The project activity is not included in any Chinese mandatory emissions trading scheme and there is no emission cap enforced for the project owner according to the compliance company list² publicly available.

Thus, the project does not reduce GHG emissions from activities that are included in an emissions trading program or any other mechanism that includes GHG allowance trading.

Other Forms of Environmental Credit

The project has not sought or received other forms of Environmental Credit, including renewable energy certificates, during this monitoring period.

1.11 Sustainable Development Contributions

The project located at the Bainitang village, Lankou Town, Dongyuan County. It is 5km from the centre of Lankou Town, 42.9km away from the centre of Heyuan City, Shenzhen, Guangdong Province, involves to employ 30 employees, and operates of 26 MW(3*8.66MW) hydro-generator units and generate power from clean renewable hydro power in Guangdong Province.

- The project has supplied 420,237.338 MWh in this monitoring period that otherwise have been produced by thermal power generation projects, and contributes to Chinese renewable energy target³.
- The project generated 30 decent working opportunities for local women and men, 15 for women and 15 for men and help them improve the incomes, and contributes to the decent work and economic growth⁴.
- The emission reduction of the project is 366,109 tCO₂e in this monitoring period, and contributes to Chinese Climate change target⁵.

¹ http://www.mee.gov.cn/xxgk2018/xxgk/xxgk05/202103/t20210330_826728.html

² <http://mee.gov.cn/xxgk2018/xxgk/xxgk03/202012/W020201230736907682380.pdf>

³ Nationally Determined Contribution Goals of renewable energy
<https://www.ccchina.org.cn/Detail.aspx?newsId=73916&TId=60>

⁴ http://www.gov.cn/zhengce/zhengceku/2021-08/27/content_5633714.htm

⁵ Nationally Determined Contribution Goals of carbon peak in 2030 and carbon neutral in 2060
<https://www.ccchina.org.cn/Detail.aspx?newsId=73916&TId=60>

Table 1: Sustainable Development Contributions

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
1)	7.2	7.2.1 Renewable energy share in the total final energy consumption	Implemented activities to increase	The project utilizes hydro energy for electricity generation, which could increase the ratio of renewable energy in total energy consumption of China. From 01/06/2012 to 31/10/2017 the electricity supplied from the project is 420,237.338MWh.	From the operation start date of the project to the end of this monitoring period, the electricity supplied from the project is around 705,250.230 MWh.

2)	8.5	Achieve full and productive employment and decent work for all women and men	Implemented activities to increase	During this monitoring period (01/06/2012 to 31/10/2017), the project achieves full and productive employment and decent work for all women and men. In 2012, there are 30 jobs created for 15 men and 15 women, the average yearly salary is 32,360RMB/year In 2013, there are 30 jobs created for 15 men and 15 women, the average yearly salary is 34,484RMB/year In 2014, there are 30 jobs created for 15 men and 15 women, the average yearly salary is 36,971RMB/year In 2015, there are 30 jobs created for 15 men and 15 women, the average yearly salary is 38,623RMB/year In 2016, there are 30 jobs created for 15 men and 15 women, the average yearly salary is 40,826RMB/year In 2017, there are 30 jobs created for 15 men and 15 women, the average yearly salary is 42,726RMB/year	From the operation start date of the project to the end of this monitoring period, the project provided 30 jobs for women and men from the year of 2007 to the year of 2017, women, and men both are worked for the project, and the average yearly salary has increased from 21,304RMB/year in 2007 to 42,726RMB/year in 2017. The project achieves full and productive employment and decent work for all women and men.
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3)	13.0	Tonnes of greenhouse gas emissions avoided or removed	Implemented activities to increase	By utilizing water for power generation and replaced it from thermal power generation projects From 01/06/2012 to 31/10/2017 the emission reduction from the project is around 366,109tCO ₂ e	From the operation start date of the project to the end of this monitoring period, the emission reduction from the project is around 614,414tCO ₂ e.
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2 SAFEGUARDS

2.1 No Net Harm

The Environmental Impact Assessment (EIA) Report was completed by Institute of Environmental Science of Zhongshan University on 12/10/2005, and approved by the Environment Protection Bureau of Guangdong province on 18/11/2005 (Document No.: Yue Huan Han [2005]1323).

According to the EIA Report, environmental impacts possibly caused by the Project and protect & guard measures adopted by the project owner are analyzed as follows:

In Construction Stage

Water

Wastewater and sewage generated by site construction activities would be treated in the sedimentation tank after collection to meet the strict national standards <Integrated wastewater discharge standard> (GB8978-1996) and then discharged, so it has less impact on Dongjiang River.

Air

The main air pollutant is particulates (dust) which is released from construction activities and transportation and the emission from vehicles and construction machinery. Measures would be taken to mitigate this pollutant, such as spraying water at construction sites and on dusty roads, transporting material in covered vehicles or in closed containers, installing and using a wheel washing system, controlling vehicle speeds, and operating with proper maintenance and in compliance with relevant emission standards.

Noise

Vehicles, construction machinery and explosion of dynamite would generate noise. The mitigation measures include: installing in-situ sound barriers, selecting suitable equipment, correct operation, and maintenance; limiting the speed of vehicles, and the explosion activities would be carried out strictly in compliance with safety regulations for explosion issued by the nation.

Solid waste

The main solid waste from this project includes: refuse generated on construction site and waste generated by construction workers. These solid wastes would be separated, collected and properly handled in approved landfills and recycling facilities.

Ecology

Measures for water and soil conservation are prepared and complied by the project owner for minimize the adverse impact on the ecological environment during the project construction. Rehabilitation of vegetation would be conducted after the construction work.

Social

No inhabitant would be relocated. The inundated area is mainly covered by small trees and bushes. The temporarily and permanently occupied lands are all barren fields.

In Operation Stage

Water

Sewage water would be treated in the treatment plant to meet the strictest discharging standards prior to be discharged.

Air

There is no air pollution caused by hydropower plant during operation stage.

Noise

Noise is generated mainly by machines during operations. The mitigation measures are: selecting low noise machines, locating noisy equipment indoors, installing noise enclosures or buffers, providing green zone with vegetation.

The project has no great adverse impact on the local people and environment.

Conclusion

To prevent the loss of soil and water during the project construction and operation particularly, the project owner also prepared an independent Scheme of Water and Soil Conservation (SWSC) for the project, which has already been approved by Water and Soil Conservation and Monitoring Bureau of Guangdong Province. According to both approved EIA and SWSC, no significant environmental impacts were identified for the project.

No resident would be relocated, no farm land and forest land to be inundated during the construction and operation period of the project. Moreover, stringent environmental monitoring would be carried out during these periods. Through these measures, the adverse impacts on environment caused by the project would be controlled and mitigated to a minimum.

2.2 Local Stakeholder Consultation

In May 2007, the project owner carried out a survey of residents, builders, and some work staffs of the Project. The survey was conducted through distributing and collecting responses to questionnaires. The survey was mainly included the project introduction, and investigation of their opinions on the Project activity as well as its impacts on the local economy and environment.

The template of questionnaire is as follows:

Questionnaire for public comments on Lankou 26MW Hydropower Project in Dongyuan County, Guangdong			
Name:	Sex:	Age:	Education:
Profession:		Address:	
Reside Time:			
Attitude to construction of the proposed project: ①Sustain ②Oppose ③Not to matter			
What positive impacts do you think construction and operation of the project have on your life? (Many can be selected)			
(1):Reduce pollutions		(5): Extend electro-range/Increase electro-quantity	
(2): Increase incomes		(6): Reduce tariff	
(3): Increase employment		(7): Improve the situation of road	
(4): Improve living standard		(8): Others (Explain generally)	
What negative impacts do you think construction of the project have on your life?			
(1): Noise		(4):Destroy local eco-system	
(2): Immigration		(5): Pollute air	
(3): Submerge		(6):Others	
farmland/Building			
What measurements do you think can reduce negative impacts? (Explain generally)			
Do you have other assessments and advices to the project? (Explain generally if have)			

Totally 47 questionnaires returned out of 50 respondents with 94% response rate. Of the 47 respondents, 26 persons are 20~29 years old, 17 persons are 31~50 years old and 4 persons are older than 50. In term of education level, 19 persons are junior high school & below level, 28 persons are senior high school & above level. The following is a summary of the key findings based on 47 returned questionnaires.

43(91.5%) persons of the respondents support the construction of the Project, 4(8.5%) persons hold a neutral attitude, and no person objects to the construction of the Project.

In term of positive impacts of the Project, the respondents consider construction and operation of the Project may improve living standard (72%); increase employment (83%); increase income (89%); extend electro-range/Increase electro-quantity (49%) and improve the situation of roads (32%).

In term of negative impacts of the Project, the respondents consider that the construction and operation of the Project could cause flood problems (70%) and lose soil & water (6.4%). It is believed those problems could be prevented by constructing protection embankment. But 91.5% respondents believe that the Project has more advantages than disadvantages.

In term of negative impacts of the Project, the respondents consider that the construction and operation of the Project could cause flood problems (70%) and lose soil & water (6.4%). It is believed those problems could be prevented by constructing protection embankment. But 91.5% respondents believe that the Project has more advantages than disadvantages.

General speaking, most local stakeholders have very positive attitudes toward to the proposed project and support its construction. They believe that the proposed project would effectively improve local economic development, social benefit and environment, and the location of the proposed project is reasonable.

The mechanism for on-going communication with local stakeholders.

The project owner has established a grievance and appeal mechanism, a grievance book is kept at the project owner office. Anyone can go there and record their grievance or comment in the book. The project owner personnel would review and respond to the comments and record the response in the book as well. Meanwhile, the local authority has also conducted spot checks on the implementation of the project from time to time as per the request from the local governments' regulations. There are no negative comments received for the project. In line with VCS requirements all the processes have been implemented to receive comments from local stakeholders as well as communicate with them at periodic intervals.

2.3 AFOLU-Specific Safeguards

As the project is a non-AFOLU project activity, therefore not applicable

3 IMPLEMENTATION STATUS

3.1 Implementation Status of the Project Activity

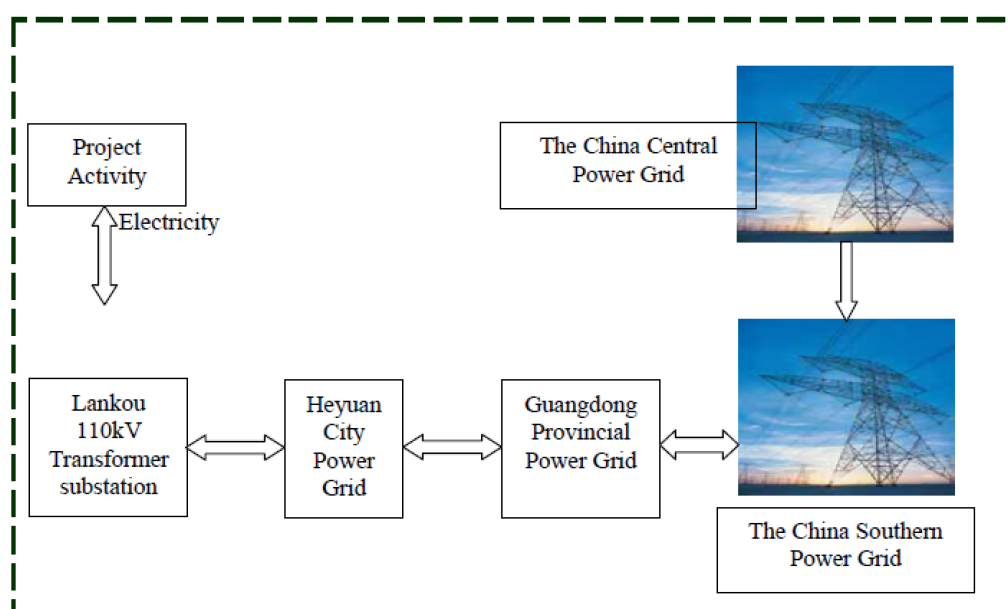
The project with total installed capacity of 26MW(3*8.66MW). The purpose of the project is to deliver electricity to China Southern Power Grid (CSPG), displacing electricity generated by thermal power plants in CSPG, thus GHG emission reduction is achieved.

The technical specifications of the turbine/generator units of the project are listed in Table 2: The key parameter of the equipment

Table 2: The key parameter of the equipment

Hydropower turbines		Generators	
Turbine Type	GZ1250a-WP-550	Type	SFWG7350-80/6220
Rated head	4.00m	Rated power	8666.7kW
Rated speed	68.2r/min	Rate factor	$\cos\phi=0.9$
Designed flow	768m ³ /s	Rated voltage	6.3kV
Lifetime	30 years	Lifetime	30 years

Figure 2 Project boundary



After commissioning, the project is operating in good conditions. During this monitoring period, the project was also running orderly. No special event occurred which may impact the GHG emission reductions and monitoring.

3.2 Deviations

3.2.1 Methodology Deviations

There are no methodology deviations applied during this monitoring period.

3.2.2 Project Description Deviations

There is a deviation of the project crediting period. The project is registered under VCS version 2007 and completed validation before 19/03/2020, thus it remains eligible to apply the crediting period requirements under VCS version 3, the first crediting period of the project should be updated from updated from 01/11/2007~06/07/2009 to 01/11/2007 - 31/10/2017 (ten years).

In the registered PDD Version 5.0 dated on 01/03/2009 in section A.4.3 Technology to be employed by the project activity table 2, Rated power of the generator is 8166.67kW, however as confirmed during the site visit, the real rated power of the generator is 8666.7kW, which is a typo. Therefore, a deviation is raised during this monitoring period.

There are no impacts on the applicability of the methodology to the project, the additionality of the project or the appropriateness of the baseline scenario to the project from the two deviations of the project description during this monitoring period.

3.3 Grouped Projects

The project is not a grouped project.

4 DATA AND PARAMETERS

4.1 Data and Parameters Available at Validation

Data / Parameter	NCVi
Data unit	TJ / mass or volume unit of a fuel
Description	The net calorific value per mass or volume unit of a fuel i
Source of data	China Energy Statistical Yearbook 2005, pg 365
Value applied	Please refer to Annex 3 of the registered PDD

Justification of choice of data or description of measurement methods and procedures applied	Official Statistical Data
Purpose of Data	Calculation of baseline emissions
Comments	Reasonable

Data / Parameter	$F_{i,j,y}$
Data unit	Mt,Mm3
Description	the amount of fuel i (in a mass or volume unit) consumed by relevant power sources j in year(s) y
Source of data	China Energy Statistical Yearbook 2005-2007
Value applied	Please refer to Annex 3 of the registered PDD
Justification of choice of data or description of measurement methods and procedures applied	Official Statistical Data
Purpose of Data	Calculation of baseline emissions
Comments	Reasonable

Data / Parameter	$GEN_{j,y}$
Data unit	MWh
Description	Electricity (MWh) delivered from each province in Southern China to the grid excluding low operating cost/must run power plants and including the electricity import from the CCPG in year y
Source of data	China Electric Power Yearbook2005-2007
Value applied	Please refer to annex 3 of the registered PDD
Justification of choice of data or description of measurement methods and procedures applied	Since the detailed generation data by power plants are not publicly available, therefore the aggregated data by fuel types are used instead.
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	$EF_{coal,adv}$
Data unit	%

Description	Efficiency of most advanced coal-fired power technology that is commercially available.
Source of data	Notice on the determination of emission factors of regional power grids by Chinese CDM DNA
Value applied	Please refer to Annex 3 of the registered PDD
Justification of choice of data or description of measurement methods and procedures applied	Official Statistical Data
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	$EF_{oil,adv}$
Data unit	%
Description	Efficiency of most advanced oil-fired power technology that is commercially available
Source of data	Notice on the determination of emission factors of regional power grids by Chinese CDM DNA
Value applied	Please refer to Annex 3 of the registered PDD
Justification of choice of data or description of measurement methods and procedures applied	Official Statistical Data
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	$EF_{gas,adv}$
Data unit	%
Description	Efficiency of most advanced gas-fired power technology that is commercially available
Source of data	Notice on the determination of emission factors of regional power grids by Chinese CDM DNA
Value applied	Please refer to Annex 3 of the registered PDD
Justification of choice of data or description of measurement methods and procedures applied	Official Statistical Data

Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	λ_i
Data unit	%
Description	The proportion of emission from different fuel i power plant to the total emissions of CSPG.
Source of data	China Electric Statistical Yearbook 2005-2007
Value applied	Please refer to Annex 3 of the registered PDD
Justification of choice of data or description of measurement methods and procedures applied	The data selected comply with “Tool to calculate the emission factor for an electricity system”.
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	$EF_{CO_2,i}$
Data unit	tCO ₂ e/TJ
Description	the CO ₂ emission factor per unit of energy of the fuel i
Source of data	IPCC 2006
Value applied	For the detailed information please see the Report on Determination of Baseline Grid Emission Factor by China DNA NDRC at http://cdm.ccchina.gov.cn .
Justification of choice of data or description of measurement methods and procedures applied	IPCC Default Value
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	$CAP_{j,y}$
Data unit	MW
Description	Installed capacity of source j in year y in Southern Power Grid
Source of data	China Electric Statistical Yearbook 2005-2007
Value applied	Please refer to Annex 3 of the registered PDD

Justification of choice of data or description of measurement methods and procedures applied	Official Statistical Data
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	Cap _{BL}
Data unit	W
Description	Install Capacity of the hydro power plant before the implementation of the project activity.
Source of data	-
Value applied	For the project is a new hydropower plant, this value is zero.
Justification of choice of data or description of measurement methods and procedures applied	Determine the installed capacity based on recognized standards.
Purpose of Data	Calculation of project emissions
Comments	-

Data / Parameter	A _{BL}
Data unit	m ²
Description	Area of the reservoir measured in the surface of the water, before the implementation of the project activity, when the reservoir is full.
Source of data	-
Value applied	For the project is a new hydropower plant, this value is zero.
Justification of choice of data or description of measurement methods and procedures applied	Measured from topographical surveys, maps, satellite pictures, etc.
Purpose of Data	Calculation of project emissions
Comments	-

4.2 Data and Parameters Monitored

Data / Parameter	EG _{output,y}				
Data unit	MWh				
Description	Electricity				
Source of data	Annual on-grid electricity supplied to CSPG by the proposed project.				
Description of measurement methods and procedures to be applied	Electricity will be measured directly and continuously by computed and precise ammeters. Recording frequency will be hourly measurement and monthly recording; proportion of data to be monitored would be 100% throughout whole year; the data will be archived both in electronic and paper; data monitored are to be kept for two years after the last issuance of CERs for the proposed project activity.				
frequency of monitoring/recording	Measuring continuously/ Recording monthly				
Value monitored	Year		Values		
	01/06/2012-31/12/2012		35,885.520		
	2013		76,430.640		
	2014		67,067.220		
	2015		61,758.180		
	2016		103,588.980		
	01/01/2017-31/10/2017		75,878.020		
	In total		420,608.560		
Monitoring equipment	M1 and M2 would be calibrated on the yearly basis				
	NO	Type	Accuracy class	Serial number	Calibration Frequency
	M1	DTSD341	0.5S	01010801048	Yearly
	M2	YD2103	0.5S	070817234	Yearly
	For detail, please refers to appendix 2.				
QA/QC procedures to be applied	Set up a special CDM project team; constitute detailed rules on monitoring management; introduce precision ammeters; two ammeters proofread each other both in booster station and switching station; keep the invoice of electricity sales as a hard proof for data quality control.				

Purpose of the data	Baseline Emission calculation
Calculation method	Measured
Comments	Double check by receipt of sales. Electronic data will be kept for 2 years following the end of the crediting period.

Data / Parameter	EG _{input,y}				
Data unit	MWh				
Description	Electricity				
Source of data	Annual on-grid electricity purchased from CSPG by the proposed project for the plant operation.				
Description of measurement methods and procedures to be applied	Electricity will be measured directly and continuously by computed and precise ammeters. Recording frequency will be hourly measurement and monthly recording; proportion of data to be monitored would be 100% throughout whole year; the data will be archived both in electronic and paper; data monitored are to be kept for two years after the last issuance of CERs for the proposed project activity.				
requency of monitoring/recording	Measuring continuously/ Recording monthly				
Value monitored	Year		Values		
	01/06/2012-31/12/2012		30.360		
	2013		62.040		
	2014		60.720		
	2015		76.520		
	2016		103.090		
	01/01/2017-31/10/2017		71.492		
	In total		404.222		
	Monitoring equipment	M1 and M2 would be calibrated on the yearly basis			
NO		Type	Accuracy class	Serial number	Calibration Frequency
M1		DTSD341	0.5S	01010801048	Yearly
M2		YD2103	0.5S	070817234	Yearly

	For detail, please refers to appendix 2.
QA/QC procedures to be applied	Set up a special CDM project team; constitute detailed rules on monitoring management; introduce precision ammeters; two ammeters proofread each other both in booster station and switching station; keep the invoice of electricity purchase as a hard proof for data quality control.
Purpose of the data	Baseline Emission calculation
Calculation method	Measured
Comments	Double check by receipt of sales. Electronic data will be kept for 2 years following the end of the crediting period.

Data / Parameter	TEG _y	
Data unit	MWh	
Description	Total electricity produced by the project activity, including the electricity supplied to the grid and the electricity supplied to internal loads, in year y.	
Source of data	Project activity site	
Description of measurement methods and procedures to be applied	Electricity will be measured directly and continuously by computed and precise ammeters. Recording frequency will be hourly measurement and monthly recording; proportion of data to be monitored would be 100% throughout whole year; the data will be archived both in electronic and paper; data monitored are to be kept for two years after the last issuance of CERs for the proposed project activity.	
Frequency of monitoring/recording	Hourly measurement and monthly recording	
Value monitored	Year	Total Value
	2012	37,076,821
	2013	78,789,527
	2014	69,162,416
	2015	63,705,721
	2016	106,693,782

	2017	78,789,527			
	In total	434,217,794			
Monitoring equipment	M3, M4 and M5 would be calibrated on the yearly basis				
	NO	Type	Accuracy class	Serial number	Calibration Frequency
	M3	YD2103	0.5S	070817236	Yearly
	M4	YD2103	0.5S	070817411	Yearly
	M5	YD2103	0.5S	070817233	Yearly
	For detail, please refers to appendix 2.				
QA/QC procedures to be applied	Set up a special CDM project team; constitute detailed rules on monitoring management; introduce precision ammeters; three ammeters installed record power generation from each generator				
Purpose of the data	Not used for Project emission, only for operational purpose				
Calculation method	-				
Comments	No comments				

Data / Parameter	Capacity
Data unit	W
Description	Installed capacity of the hydro power plant after the implementation of the project activity.
Source of data	Project site
Description of measurement methods and procedures to be applied	Verified on site
Frequency of monitoring/recording	Yearly
Value monitored	260,000,000
Monitoring equipment	-
QA/QC procedures to be applied	-

Purpose of the data	Calculation of project emissions
Calculation method	-
Comments	No comments

Data / Parameter	A _{PJ}															
Data unit	m ²															
Description	The area of the reservoir measured in the surface of the water, after the implementation of the project activity, when the reservoir is full															
Source of data	Measured															
Description of measurement methods and procedures to be applied	The surface area will be calculated using the design schematics and area maps. Photographs of the reservoir at several key locations will be taken when the project becomes operational to check whether the actual reservoir does not deviate substantially for the design.															
Frequency of monitoring/recording	Yearly															
Value monitored	<table><tr><td>Year</td><td>A_{PJ}(m²)</td></tr><tr><td>2012</td><td>141,441</td></tr><tr><td>2013</td><td>141,438</td></tr><tr><td>2014</td><td>141,440</td></tr><tr><td>2015</td><td>141,441</td></tr><tr><td>2016</td><td>141,443</td></tr><tr><td>2017</td><td>141,441</td></tr></table>		Year	A _{PJ} (m ²)	2012	141,441	2013	141,438	2014	141,440	2015	141,441	2016	141,443	2017	141,441
Year	A _{PJ} (m ²)															
2012	141,441															
2013	141,438															
2014	141,440															
2015	141,441															
2016	141,443															
2017	141,441															
Monitoring equipment	-															
QA/QC procedures to be applied	-															
Purpose of the data	Calculation of project emissions															

Calculation method	-
Comments	No comments

4.3 Monitoring Plan

1. Organization structure

Prior to the start of the crediting period, the organization of the monitoring team would be established. Clear roles and responsibilities would be assigned to all staff involved in the CDM project and a single CDM Department Manager would be nominated. The CDM Department Manager would have the overall responsibility for the monitoring system on this project.

The structure of the monitoring organization is shown in the figure 3 below

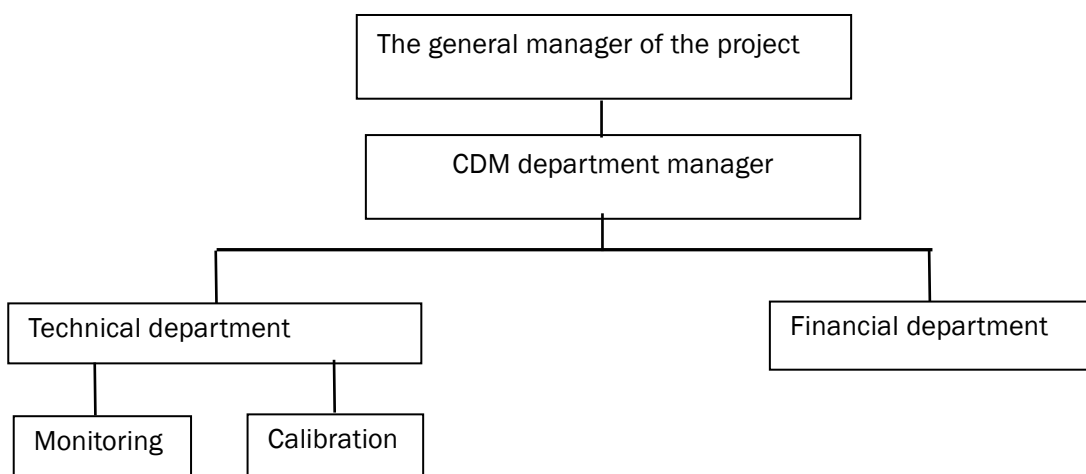


Figure 3: The structure of the monitoring organization

All other CDM monitoring staff would have clearly defined roles and responsibilities. The CDM Department Manager would manage the process of training new staff, ensuring trained staffs perform the monitoring duties and that where trained monitoring staffs are absent, and the integrity of the monitoring system is maintained by other trained staff.

A formal set of monitoring procedures would be established prior to the start of the project. These procedures would detail the organization, control and steps required for certain key monitoring system features, including:

- 1) CDM staff training
- 2) Data management systems
- 3) CDM data QA/QC

- 4) Electricity meter maintenance/calibration
- 5) Equipment failure

The CDM Department Manager would be responsible for ensuring that the procedures are followed on site and for continuously improving the procedures to ensure a reliable monitoring system is established.

All staff involved in the CDM project would receive some relevant training from the project consulting company laid down in training procedures agreed on by the project developer and consulting company. Records of trained CDM staff would be retained by the Project Developer. The CDM Department Manager would ensure that only trained staffs are involved in the operation of the monitoring system.

2. Monitoring data

The main monitoring data are:

- 1) Annual on-grid electricity supplied to CSPG by the proposed project $EG_{output,y}$; and
- 2) Annual on-grid electricity purchased from CSPG by the proposed project for the plant operation $EG_{input,y}$; and
- 3) Annual Installed Capacity of the project Cap_P ;
- 4) Annual Reservoir Area after the project implementation (A_P); and
- 5) Total electricity produced by the project activity, including the electricity supplied to the grid and the electricity supplied to internal loads TEG_y .

Since the baseline emission factor is fixed by Ex-ante calculation, the net grid-connected electricity generation by the proposed project supplied to CSPG EG_y is calculated by difference of 1) and 2) above, that is $EG_y = EG_{output,y} - EG_{input,y}$.

3. Monitoring equipment and installation

The electric energy metering equipment would be properly configured, and the metering equipment would be checked by both the project owner and the grid company before the project is in operation. Periodic calibrates should be conducted according to the relevant national standard

Main meter (M1) is employed at the entrance of the Lankou 110kV Transformer Station to measure annual on-grid electricity supplied to CSPG by the proposed project $EG_{output,y}$ and annual on-grid electricity purchased from CSPG by the proposed project for the plant operation $EG_{input,y}$.

Backup meter (M2) is employed at the exit of the booster station of the proposed project to measure output electricity of the project.

Backup meters (M3、 M4 and M5) are employed at the exits of the generators of the project to measure the total electricity produced by the project activity TEGy.

Through the main meter M1 (accuracy degree is 0.5S, two-direction), the amount of on-grid electricity supplied to CSPG by the proposed project $EG_{output,y}$ and on-grid electricity purchased from CSPG by the proposed project for the plant operation $EG_{input,y}$ can be confirmed.

The net grid-connected electricity generation by the proposed project supplied to CSPG EG_y would be calculated as: $EG_{output,y} - EG_{input,y}$.

A schematic diagram to describe the wire connection has been shown in the Fig 4

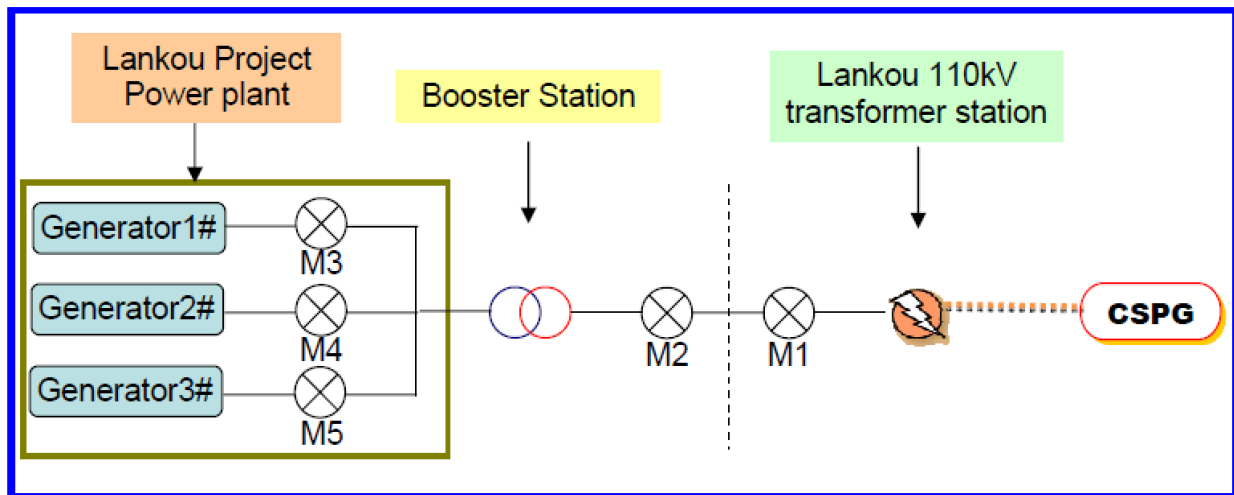


Figure 4: the Wire Connection Diagram of the Lankou project

4. Data Collection

The process for collecting the electricity meter data would be detailed in a procedure. A summary of this procedure is provided below.

Data of meters

The project owner and the Grid Company are responsible for operation monitoring of backup meters and main meter respectively, they should guarantee the measure equipment favorable with integrate seal.

The main monitoring processes are as follows:

- i. The project owner and Grid Company read and check the backup meter and the main meter and record the data on the stated day of every month;
- ii. The grid company delivers the electricity supplied by the grid to the project owner;

- iii. The project owner provides an electricity sales invoice to the Grid Company. A copy of the invoice is stored by the project owner, together with a record of the payment by the grid company;
- iv. The Grid Company provides an electricity purchases invoice to the project owner and the invoice is stored by the project owner;
- v. The project owner records the net grid-connected electricity generation by the proposed project supplied to CSPG EGy;
- vi. The project owner keeps and safeguards the records of the main meter's data readings for verification by the DOE.

An overview of the recording frequency, calibration procedures and available documentation is provided in Table 3. The numbering of the metering equipment refers to Figure 4 which shows the location of each meter.

Table 3 Details of metering equipment

Meter	Recorded by	Frequency	Accuracy	Period of calibration	Documentation
M1	Grid company and Project owner	Hourly measurement and monthly recording	0.5S	12 months	Print out of electronic record and optional paper log
M2	Project owner		0.5S		
M3			0.5S		
M4			0.5S		
M5			0.5S		

Damages to metering equipment:

In case metering equipment is damaged and no reliable readings can be recorded, the Project Owner would estimate the net grid-connected electricity generation by the proposed project supplied to CSPG EGy according to the following procedure:

1) In case metering equipment (M1) operated by Grid Company is damaged only:

The meter (M2) data logged by the Project Owner, evidenced by electronic records would be used as record of power supplied to the grid for the days for which no record could be recorded. The

line-loss would be calculated according to the data before. The meters (M3, M4 and M5) are still used as record of total electricity produced by the project activity.

2) In case metering equipment (M1 and M2) operated by Grid Company is damaged:

The meters (M3, M4 and M5) data logged by the Project Owner, evidenced by electronic records would be used as record of power supplied to the grid and total electricity produced by the project activity for the days for which no record could be recorded. The line-loss would be calculated according to the data before.

3) In case metering equipment (M3, M4 and M5) operated by Grid Company is damaged:

The meter (M2) data logged by the Project Owner, evidenced by electronic records would be used as record of total electricity produced by the project activity for the days for which no record could be recorded. The line-loss would be calculated according to the data before.

Emergencies:

In case of emergencies, the Project Owner would not claim emission reductions due to the project activity for the duration of the emergency. The Project Owner would follow the following procedure for declaring the emergency period to be over:

- 1) The Project Owner would ensure that all requirements for monitoring of emission reductions have been re-established.
- 2) The monitoring officer and the head of operations of the hydropower station would both sign a statement declaring the emergency to have ended and normal operations to have resumed.

5. Data and records management

At the end of each month the monitoring data needs to be filed electronically. The electronic files need to have CD back-up or print-out. The project developer needs to keep electricity sale and purchase invoices.

All written documentation such as maps, drawings, the EIA and the Feasibility study, should be stored and should be available to the verifier so that the reliability of the information may be checked.

In order to make it easy for the verifier to retrieve the documentation and information in relation to the project emission reduction verification, the project developer should provide a document register. The document management system would be developed to ensure adequate document control for CDM purposes.

The dedicated CDM Department Manager of the project developer is responsible for checking the data (according to a formal procedure) and the CDM Department Manager would be responsible for managing the collection, storage and archive of all data and records. A procedure would be developed to manage the CDM record keeping arrangements. All the data shall be kept for two years after the end of the crediting period or the last issuance of CERs, for this project activity, whichever occurs later.

5 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

Baseline emissions include only CO₂ emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity, calculated as follows:

$$BE_y = (EG_y - EG_{\text{baseline}}) \times EF_{\text{grid,CM,y}} \quad (1)$$

Where:

BE_y = Baseline emissions in year y (tCO₂e/y);

EG_y = Net electricity supplied by the project activity to the grid (MWh);

EG_{baseline} = Baseline electricity supplied to the grid in the case of modified or retrofit facilities (MWh). For new power plants this value is taken as zero;

$EF_{\text{grid,CM,y}}$ = Combined margin CO₂ emission factor for grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system” (tCO₂e/MWh) .

Since this project installs a new grid-connected renewable power plant/unit, $EG_{\text{baseline}} = 0$; therefore, $BE_y = EG_y \times EF_{\text{grid,CM,y}}$.

In accordance with the above stated monitoring parameters in section 4.3, for the project activity:

$$EG_y = EG_{\text{output,y}} - EG_{\text{input,y}} \quad (2)$$

$EG_{\text{output,y}}$ is annual on-grid electricity supplied to CSPG by the proposed project in year y

$EG_{\text{input,y}}$ is annual on-grid electricity purchased from CSPG by the proposed project for the plant operation in year y.

The baseline emissions calculations are shown below in Table 4 and Appendix 3.

Baseline emissions calculations

The annually and monthly baseline emissions calculations are shown in Table 4 and Appendix 3.

Table 4 Annually Baseline emission calculation

Period	EG _y (MWh)	EF _{grid,CM, y} (tCO _{2e} /MWh)	BE _y (tCO _{2e})
	A	B	C=Rounddown(A X B)
01/06/2012-31/12/2012	35,855.160	0.8712	31,237
01/01/2013-31/12/2013	76,401.600	0.8712	66,561
01/01/2014-31/12/2014	67,006.500	0.8712	58,376
01/01/2015-31/12/2015	61,681.660	0.8712	53,737
01/01/2016-31/12/2016	103,485.890	0.8712	90,156
01/01/2017-31/10/2017	75,806.528	0.8712	66,042

5.2 Project Emissions

As per ACM0002 version 7, the project emissions of the project equal to the emissions from the reservoir.

The power density of the project in each year of this monitoring period is listed in Table 5 based on the installed capacity of the hydropower plant and the surface area of the reservoir.

Table 5. The power density of the project in this monitoring period

Year	Cap _{PI} (W)	A _{PI} (m ²)	Power Density (W/m ²)
	A	B	C=A/B
2012	26,000,000	141,441	183.822
2013	26,000,000	141,438	183.826
2014	26,000,000	141,440	183.824
2015	26,000,000	141,441	183.822
2016	26,000,000	141,443	183.820
2017	26,000,000	141,441	183.822

The power density in each year of this monitoring period is above the threshold of 10 W/m², as per the methodology ACM0002 version 7, the emissions from the reservoir are neglected.

Therefore, the emission of the project is 0.

5.3 Leakage

No leakage is considered in accordance with ACM0002 version 07.

5.4 Net GHG Emission Reductions and Removals

This monitoring period started from 01/06/2012 and ended at 31/10/2017 (first and last days included), with totally 1,979 days, based on the annual estimated emission reductions from the registered PDD, the amount of emission reductions from this monitoring period would be 69,757tCO₂e/365*1,979 days=378,216tCO₂e. The actual emission reductions in this monitoring period are 366,109tCO₂e, which is 3.2% lower than the estimated ERs.

Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals (tCO ₂ e)
01/06/2012-31/12/2012	31,237	0	0	31,237
01/01/2013-31/12/2013	66,561	0	0	66,561
01/01/2014-31/12/2014	58,376	0	0	58,376
01/01/2015-31/12/2015	53,737	0	0	53,737
01/01/2016-31/12/2016	90,156	0	0	90,156
01/01/2017-31/10/2017	66,042	0	0	66,042
Total	366,109	0	0	366,109

APPENDIX 1 <EVIDENCES OF SUSTAINABLE DEVELOPMENT GOAL>

Evidence for SDG 8.5



2007 年广东蓝口 26MW 水电项目年工资单
Staff name-list and salary of the project in 2007

序号 NO	姓名 Name	性别 Gender	年工资 Annual salary
1	缪永康	男	52340.00
2	张展文	男	74072.80
3	曾顺	男	4076.00
4	缪国强	男	38219.08
5	张育辉	男	23699.08
6	陈元龙	男	30454.08
7	罗国雄	男	28499.08
8	陈德华	男	88890.68
9	曹佳源	男	86284.24
10	杨兴湘	男	64120.00
11	汪文良	男	83012.00
12	陈汉良	男	27033.56
13	何建东	男	32436.00
14	刘丽燕	女	22593.56
15	刘雪红	女	4600.00
16	邹华英	女	12148.56
17	李卫华	女	22553.56
18	曾庆科	女	21233.56
19	张石胜	女	21983.56
20	曾宪威	女	20273.56
21	曾宏新	女	21700.56
22	缪乃清	女	12453.56
23	李信华	女	18293.56
24	曾庆扬	女	13233.56
25	曾建清	女	17823.56
26	杨上礼	女	12065.56
27	张启强	男	13133.56
28	陈伶	女	13010.56
29	廖国强	男	22433.56

30	曾恒忠	女	22193.56
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2012年广东蓝口26MW水电项目年工资单
Staff name-list and salary of the project in 2012

序号 NO	姓名 Name	性别 Gender	年工资 Annual salary
1	缪永康	男	56550.00
2	张展文	男	79072.80
3	曾顺	男	4276.00
4	缪国强	男	40219.08
5	张育辉	男	25699.08
6	陈元龙	男	34759.08
7	罗国雄	男	30499.08
8	陈德华	男	109990.68
9	曹佳源	男	88254.24
10	杨兴湘	男	70172.00
11	汪文良	男	93012.00
12	陈汉良	男	31033.56
13	何建东	男	35436.00
14	刘丽燕	女	23593.56
15	刘雪红	女	4800.00
16	邹华英	女	12253.56
17	李卫华	女	24553.56
18	曾庆科	女	23233.56
19	张石胜	女	22993.56
20	曾宪威	女	22273.56
21	曾宏新	女	21985.56
22	缪乃清	女	14953.56
23	李信华	女	19393.56
24	曾庆扬	女	13633.56
25	曾建清	女	18313.56
26	杨上礼	女	12265.56

27	张启强	男	13633.56
28	陈伶	女	13633.56
29	廖国强	男	23233.56
30	曾恒忠	女	22993.56

2013年广东蓝口26MW水电项目年工资单
Staff name-list and salary of the project in 2013

序号 NO	姓名 Name	性别 Gender	年工资 Annual salary
1	缪永康	男	56252.28
2	张展文	男	78299.76
3	曾顺	男	51244.08
4	缪国强	男	40153.92
5	张育辉	男	27229.92
6	陈元龙	男	31753.92
7	罗国雄	男	33193.92
8	陈德华	男	101319.84
9	曹佳源	男	87112.80
10	汪文良	男	93204.00
11	陈汉良	男	34085.28
12	何建东	男	58926.48
13	刘丽燕	女	25925.28
14	李卫华	男	23525.28
15	曾庆科	女	24725.28
16	张石胜	女	22085.28
17	曾宏新	女	22325.28
18	缪乃清	女	22325.28
19	李信华	女	19685.28
20	曾庆扬	女	16805.28
21	林志舜	女	15605.28
22	刘雪红	女	14645.28
23	邹华英	女	14285.28
24	曾建清	女	20645.28

25	杨上礼	女	15845.28
26	张启强	男	17045.28
27	陈伶	女	16805.28
28	廖国强	男	22325.28
29	曾恒忠	女	22085.28
30	叶京娣	女	15965.28



2014年广东蓝湾26MW 水电项目年工资单

Staff name-list and salary of the project in 2014

序号 NO	姓名 Name	性别 Gender	年工资 Annual salary
1	缪永康	男	59878.44
2	张展文	男	81694.44
3	曾顺	男	57064.08
4	缪国强	男	46037.16
5	张育辉	男	33238.08
6	张伟明	男	42312.36
7	陈元龙	男	37522.08
8	罗国雄	男	37762.08
9	陈德华	男	112653.60
10	曹佳源	男	92301.60
11	汪文良	男	98196.00
12	李冰	男	30552.00
13	陈汉良	男	39672.00
14	何建东	男	64580.88
15	刘丽燕	女	31752.00
16	李卫华	男	29352.00
17	曾庆科	女	30552.00
18	张石胜	女	27912.00

19	曾宏新	女	28152.00
20	缪乃清	女	28152.00
21	李信华	女	25512.00
22	曾庆扬	女	22632.00
23	林志舜	女	21432.00
24	刘雪红	女	20712.00
25	邹华英	女	20112.00
26	曾建清	女	26232.00
27	周运新	女	15192.00
28	陈伶	女	22632.00
29	叶京娣	女	21792.00
30	叶群英	女	19872.00



2015 年广东蓝口 26MW 水电项目年工资单

Staff name-list and salary of the project in 2015

序号 NO	姓名 Name	性别 Gender	年工资 Annual salary
1	缪永康	男	60,094.44
2	张展文	男	81,910.44
3	曾顺	男	57,211.80
4	缪国强	男	46,037.40
5	张育辉	男	33,238.32
6	张伟明	男	42,312.60
7	陈元龙	男	37,282.32
8	罗国雄	男	37,522.32
9	陈德华	男	95,892.00
10	曹佳源	男	86,907.96
11	汪文良	男	98,388.00
12	李冰	男	30,359.28
13	陈汉良	男	38,550.00
14	何建东	男	64,718.04
15	李卫华	男	29,217.12
16	曾庆科	女	30,119.28
17	张石胜	女	27,777.12
18	曾宏新	女	28,017.12
19	缪乃清	女	16,017.12

20	李信华	女	25,377.12
21	曾庆扬	女	22,497.12
22	林志舜	女	21,297.12
23	刘雪红	女	20,337.12
24	邹华英	女	19,977.12
25	曾建清	女	2,194.76
26	周运新	女	21,057.12
27	张启强	男	22,737.12
28	陈伶	女	22,497.12
29	叶京娣	女	21,657.12
30	叶群英	女	19,737.12

2016 年广东蓝口 26MW 水电项目年工资单

Staff name-list and salary of the project in 2016

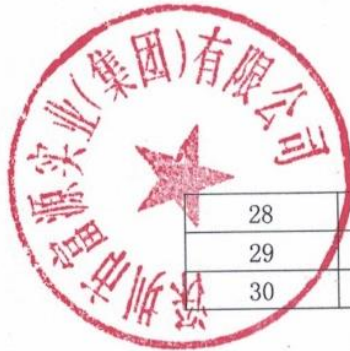
序号 NO	姓名 Name	性别 Gender	年工资 Annual salary
1	缪永康	男	68,355.96
2	张展文	男	82,126.44
3	曾顺	男	62,990.04
4	缪国强	男	30,600.00
5	张伟明	男	48,365.40
6	陈元龙	男	39,682.32
7	罗国雄	男	43,476.60
8	缪伯乐	男	36,682.32
9	廖孟良	男	41,842.32
10	谢明华	男	52,707.12
11	曾建祥	男	24,417.48
12	缪佰展	男	45,339.00
13	陈德华	男	105,684.00
14	曹佳源	男	97,458.24
15	李冰	男	30,456.48
16	陈汉良	男	48,087.72
17	何建东	男	70,334.04
18	李卫华	女	31,943.76
19	曾庆科	女	32,856.48
20	张石胜	女	30,503.76
21	曾宏新	女	30,743.76
22	曾恒忠	女	30,503.76
23	李信华	女	28,103.76
24	曾庆扬	女	25,223.76

25	林志舜	女	24,023.76
26	刘雪红	女	23,063.76
27	邹华英	女	22,703.76
28	曾建清	女	28,823.76
29	周运新	女	23,783.76
30	陈伶	女	25,223.76

2017年广东蓝口26MW水电项目年工资单

Staff name-list and salary of the project in 2017

序号 NO	姓名 Name	性别 Gender	年工资 Annual salary
1	缪永康	男	68,571.96
2	张展文	男	82,342.44
3	曾顺	男	63,206.04
4	缪国强	男	30,600.00
5	张伟明	男	48,598.20
6	陈元龙	男	39,682.32
7	罗国雄	男	43,476.60
8	缪伯乐	男	32,602.32
9	廖孟良	男	37,762.32
10	谢明华	男	52,939.92
11	曾建祥	男	24,657.48
12	缪佰展	男	45,571.80
13	陈德华	男	105,876.00
14	邱国城	男	159,216.00
15	曹佳源	男	97,650.24
16	陈汉良	女	48,320.52
17	何建东	男	69,897.24
18	李卫华	女	27,934.80
19	曾庆科	女	28,855.20
20	张石胜	女	26,494.80
21	曾宏新	女	26,734.80
22	曾恒忠	女	26,494.80
23	李信华	女	28,414.80
24	曾庆扬	女	25,534.80
25	林志舜	女	24,334.80
26	刘雪红	女	23,374.80
27	邹华英	女	23,014.80



28	曾建清	女	29,134.80
29	周运新	女	24,094.80
30	陈伶	女	25,534.80

Evidence for SDG7.2 and 13.0

The electricity supplied during this monitoring period between 01/06/2012 and 31/10/2017 is 420,806.862MWh. The total emission reductions claimed in this monitoring period are 366,109tCO_{2e}. and the emission reduction generated from the project during this monitoring period is 366,109tCO_{2e} as mentioned in section 5.4 above, which is calculated from meter(s) readings and ETNs provided by the project owner, which would be verified by VVB.

APPENDIX 2 THE CALIBRATION RECORDS OF ELECTRICITY METERS

Meter	Serial Number	Type	Accuracy	Calibration date	Valid until	Qualified third party
M1	01010801048	DTSD341	0.5S	12/04/2012	11/04/2013	Heyuan Power Supply Bureau
				11/04/2013	10/04/2014	
				10/04/2014	09/04/2015	
				09/04/2015	08/04/2016	
				08/04/2016	07/04/2017	
				07/04/2017	06/04/2018	
M2	070817234	YD2103	0.5S	12/04/2012	11/04/2013	
				11/04/2013	10/04/2014	
				10/04/2014	09/04/2015	
				09/04/2015	08/04/2016	
				08/04/2016	07/04/2017	
				07/04/2017	06/04/2018	
M3	070817236	YD2103	0.5S	12/04/2012	11/04/2013	
				11/04/2013	10/04/2014	
				10/04/2014	09/04/2015	
				09/04/2015	08/04/2016	
				08/04/2016	07/04/2017	
				07/04/2017	06/04/2018	
M4	070817411	YD2103	0.5S	12/04/2012	11/04/2013	
				11/04/2013	10/04/2014	
				10/04/2014	09/04/2015	
				09/04/2015	08/04/2016	
				08/04/2016	07/04/2017	
				07/04/2017	06/04/2018	
M5	070817233	YD2103	0.5S	12/04/2012	11/04/2013	
				11/04/2013	10/04/2014	
				10/04/2014	09/04/2015	
				09/04/2015	08/04/2016	
				08/04/2016	07/04/2017	
				07/04/2017	06/04/2018	

APPENDIX 3 < MONTHLY BASELINE EMISSIONS CALCULATIONS >

Monitoring Period	Meter reading		ETNs		Cross check between EG _{output,y} (Electricity supplied to the grid) and ETNs Electricity exported to the Grid (MWh)	Max (EG _{input,y} (Electricity consumption from the grid, ETNs Electricity consumption from the Grid) (MWh)	EG (MWh)	EF _{grid,CM, y} (tCO ₂ e/MWh)	Baseline emissions (tCO ₂ e)
	EG _{output,y} (Electricity supplied to the grid) MWh	EG _{input,y} (Electricity consumption from the grid) MWh	Electricity exported to the Grid (MWh)	Electricity consumption from the Grid (MWh)					
	A	B	C	D					
01/06/2012 to 30/06 ⁶ /2012	8625.54	0	8,625.540	0	8,625.540	0	8,625.540	0.8712	7,514.6
01/07/2012 to 31/07/2012	6148.56	1.32	6,148.560	1.32	6,148.560	1.32	6,147.240	0.8712	5,355.5
01/08/2012 to 31/08/2012	5616.6	1.98	5,616.600	1.98	5,616.600	1.98	5,614.620	0.8712	4,891.5
01/09/2012 to 30/09/2012	3423.42	6.6	3,423.420	6.6	3,423.420	6.6	3,416.820	0.8712	2,976.7
01/10/2012 to 31/10/2012	3212.88	10.56	3,212.880	10.56	3,212.880	10.56	3,202.320	0.8712	2,789.9
01/11/2012 to 30/11/2012	3896.64	7.26	3,896.640	7.26	3,896.640	7.26	3,889.380	0.8712	3,388.4
01/12/2012 to 31/12/2012	4961.88	2.64	4,961.880	2.64	4,961.880	2.64	4,959.240	0.8712	4,320.5
Subtotal of 2012	35885.52	30.36	35,885.520	30.36	35885.52	30.36	35,855.160	-	31,237 ⁷
01/01/2013 to 31/01/2013	4757.94	4.62	4,757.940	4.62	4,757.940	4.62	4,753.320	0.8712	4,141.1
01/02/2013 to 28/02/2013	3991.68	3.3	3,991.680	3.3	3,991.680	3.3	3,988.380	0.8712	3,474.7
01/03/2013 to 31/03/2013	5474.04	0.66	5,474.040	0.66	5,474.040	0.66	5,473.380	0.8712	4,768.4

⁶ The grid company and the plant operation staff read and record the meters at 24:00 of the last day of each month separately. The recording of the exported and imported electricity is at 24:00 of the last day of each month separately during period from 01/06/2012 to 31/10/2017.

⁷ Only the annually baseline emissions round-down, the monthly baseline emissions do not round-down.

01/04/2013 to 30/04/2013	6064.74	1.98	6,064.740	1.98	6,064.740	1.98	6,062.760	0.8712	5,281.9
01/05/2013 to 31/05/2013	7716.06	3.3	7,716.060	3.3	7,716.060	3.3	7,712.760	0.8712	6,719.4
01/06/2013 to 30/06/2013	11239.8	0	11,239.800	0	11,239.800	0	11,239.800	0.8712	9,792.1
01/07/2013 to 31/07/2013	8939.04	0	8,939.040	0	8,939.040	0	8,939.040	0.8712	7,787.7
01/08/2013 to 31/08/2013	9298.74	0.66	9,298.740	0.66	9,298.740	0.66	9,298.080	0.8712	8,100.5
01/09/2013 to 30/09/2013	6933.96	0.66	6,933.960	0.66	6,933.960	0.66	6,933.300	0.8712	6,040.3
01/10/2013 to 31/10/2013	5103.12	3.96	5,103.120	3.96	5,103.120	3.96	5,099.160	0.8712	4,442.4
01/11/2013 to 30/11/2013	3934.26	5.28	3,934.260	5.28	3,934.260	5.28	3,928.980	0.8712	3,422.9
01/12/2013 to 31/12/2013	2977.26	4.62	2,977.260	4.62	2,977.260	4.62	2,972.640	0.8712	2,589.8
Subtotal of 2013	76430.64	62.04	76430.64	29.04	76,430.640	62.04	76,401.600	-	66,561
01/01/2014 to 31/01/2014	6343.92	9.24	6,343.920	9.24	6,343.920	9.24	6,334.680	0.8712	5,518.8
01/02/2014 to 28/02/2014	4320.36	0.66	4,320.360	0.66	4,320.360	0.66	4,319.700	0.8712	3,763.3
01/03/2014 to 31/03/2014	6328.74	1.32	6,328.740	1.32	6,328.740	1.32	6,327.420	0.8712	5,512.4
01/04/2014 to 30/04/2014	5471.4	3.3	5,471.400	3.3	5,471.400	3.3	5,468.100	0.8712	4,763.8
01/05/2014 to 31/05/2014	9139.68	0.66	9,139.680	0.66	9,139.680	0.66	9,139.020	0.8712	7,961.9
01/06/2014 to 30/06/2014	9385.2	0.66	9,385.200	0.66	9,385.200	0.66	9,384.540	0.8712	8,175.8
01/07/2014 to 31/07/2014	6559.08	1.98	6,559.080	1.98	6,559.080	1.98	6,557.100	0.8712	5,712.5
01/08/2014 to 31/08/2014	5351.94	5.28	5,351.940	5.28	5,351.940	5.28	5,346.660	0.8712	4,658.0
01/09/2014 to 30/09/2014	4210.8	9.24	4,210.800	9.24	4,210.800	9.24	4,201.560	0.8712	3,660.4
01/10/2014 to 31/10/2014	3110.58	10.56	3,110.580	10.56	3,110.580	10.56	3,100.020	0.8712	2,700.7
01/11/2014 to 30/11/2014	3245.88	9.24	3,245.880	9.24	3,245.880	9.24	3,236.640	0.8712	2,819.8
01/12/2014 to 31/12/2014	3599.64	8.58	3,599.640	8.58	3,599.640	8.58	3,591.060	0.8712	3,128.5
Subtotal of 2014	67067.22	60.72	67067.22	60.72	67067.22	60.72	67006.5	-	58,376
01/01/2015 to 31/01/2015	4328.28	7.26	4,328.280	7.26	4,328.280	7.26	4,321.020	0.8712	3,764.5
01/02/2015 to 28/02/2015	5000.16	7.92	5,000.160	7.92	5,000.160	7.92	4,992.240	0.8712	4,349.2
01/03/2015 to 31/03/2015	984.72	13.2	984.720	13.2	984.720	13.2	971.520	0.8712	846.4

01/04/2015 to 30/04/2015	2612.94	12.54	2,612.940	12.54	2,612.940	12.54	2,600.400	0.8712	2,265.5
01/05/2015 to 31/05/2015	7113.48	1.98	7,113.480	1.98	7,113.480	1.98	7,111.500	0.8712	6,195.5
01/06/2015 to 30/06/2015	7704.18	0.66	7,704.180	0.66	7,704.180	0.66	7,703.520	0.8712	6,711.3
01/07/2015 to 31/07/2015	6654.12	5.94	6,654.120	5.94	6,654.120	5.94	6,648.180	0.8712	5,791.9
01/08/2015 to 31/08/2015	6807.9	3.3	6,807.900	3.3	6,807.900	3.3	6,804.600	0.8712	5,928.2
01/09/2015 to 30/09/2015	6091.342	7.462	6,083.880	7.462	6,083.880	7.462	6,076.418	0.8712	5,293.8
01/10/2015 to 31/10/2015	4559.332	6.652	4,552.680	6.652	4,552.680	6.652	4,546.028	0.8712	3,960.5
01/11/2015 to 30/11/2015	3419.888	3.728	3,416.160	3.728	3,416.160	3.728	3,412.432	0.8712	2,972.9
01/12/2015 to 31/12/2015	6505.558	5.878	6,499.680	5.878	6,499.680	5.878	6,493.802	0.8712	5,657.4
Subtotal of 2015	61781.9	76.52	61758.18	76.52	61758.18	76.52	61681.66	-	53,737
01/01/2016 to 31/01/2016	6629.224	6.784	6,622.440	6.784	6,622.440	6.784	6,615.656	0.8712	5,763.6
01/02/2016 to 28/02/2016	10099.444	9.364	10,090.080	9.364	10,090.080	9.364	10,080.716	0.8712	8,782.3
01/03/2016 to 31/03/2016	8616.988	8.608	8,608.380	8.608	8,608.380	8.608	8,599.772	0.8712	7,492.1
01/04/2016 to 30/04/2016	8149.231	8.791	8,140.440	8.791	8,140.440	8.791	8,131.649	0.8712	7,084.3
01/05/2016 to 31/05/2016	10441.111	10.471	10,430.640	10.471	10,430.640	10.471	10,420.169	0.8712	9,078.1
01/06/2016 to 30/06/2016	10142.07	11.73	10,130.340	11.73	10,130.340	11.73	10,118.610	0.8712	8,815.3
01/07/2016 to 31/07/2016	10109.812	9.832	10,099.980	9.832	10,099.980	9.832	10,090.148	0.8712	8,790.5
01/08/2016 to 31/08/2016	8138.562	8.682	8,129.880	8.682	8,129.880	8.682	8,121.198	0.8712	7,075.2
01/09/2016 to 30/09/2016	8906.11	8.65	8,897.460	8.65	8,897.460	8.65	8,888.810	0.8712	7,743.9
01/10/2016 to 31/10/2016	7956.99	7.29	7,949.700	7.29	7,949.700	7.29	7,942.410	0.8712	6,919.4
01/11/2016 to 30/11/2016	7475.899	6.019	7,469.880	6.019	7,469.880	6.019	7,463.861	0.8712	6,502.5
01/12/2016 to 31/12/2016	7026.629	6.869	7,019.760	6.869	7,019.760	6.869	7,012.891	0.8712	6,109.6
Subtotal of 2016	103692.07	103.09	103,588.980	103.09	103,588.98	103.09	103,485.89	-	90,156
01/01/2017 to 31/01/2017	8949.778	8.758	8,941.020	8.758	8,941.020	8.758	8,932.262	0.8712	7,781.8
01/02/2017 to 28/02/2017	5219.782	5.122	5,214.660	5.122	5,214.660	5.122	5,209.538	0.8712	4,538.5
01/03/2017 to 31/03/2017	6958.37	5.93	6,952.440	5.93	6,952.440	5.93	6,946.510	0.8712	6,051.8

01/04/2017 to 30/04/2017	6771.73	5.41	6,766.320	5.41	6,766.320	5.41	6,760.910	0.8712	5,890.1
01/05/2017 to 31/05/2017	7863.59	6.95	7,856.640	6.95	7,856.640	6.95	7,849.690	0.8712	6,838.6
01/06/2017 to 30/06/2017	10553.836	9.676	10,544.160	9.676	10,544.160	9.676	10,534.484	0.8712	9,177.6
01/07/2017 to 31/07/2017	10774.474	10.074	10,764.400	10.074	10,764.400	10.074	10,754.326	0.8712	9,369.2
01/08/2017 to 31/08/2017	7298.66	7.64	7,291.020	7.64	7,291.020	7.64	7,283.380	0.8712	6,345.3
01/09/2017 to 30/09/2017	6016.23	6.93	6,009.300	6.93	6,009.300	6.93	6,002.370	0.8712	5,229.3
01/10/2017 to 31/10/2017	5543.062	5.002	5,538.060	5.002	5,538.060	5.002	5,533.058	0.8712	4,820.4
Subtotal of 01/01/2017-31/10/2017	75949.512	71.492	75878.02	71.492	75878.02	71.492	75806.528	-	66,042
In total	420806.862	404.222	420608.560	371.222	420608.560	404.222	420237.338	-	366,109